HVAC HOUSING

DESCRIPTION

Both the single zone manual temperature control (MTC) heating-A/C system and the heater-only system are equipped with a common HVAC housing assembly that combines A/C and heating capabilities into a single unit located within the passenger compartment (Fig. 7). The HVAC housing assembly consists of the following:

- **HVAC housing** — The HVAC housing consists of an upper housing and a lower housing that are attached together by screws and clips and is mounted to the dash panel behind the instrument panel. The HVAC housing contains the heater core, A/C evaporator, blend-air and mode-air doors, and has mounting provisions for the air-door actuators, floor distribution duct, blower motor, blower motor resistor, blower motor relay and the HVAC wire harness and vacuum harness.
- **Air inlet housing** — The air inlet housing is mounted to the right end of the HVAC housing and contains the recirculation-air door and has mounting provisions for the recirculation door actuator and its linkage.

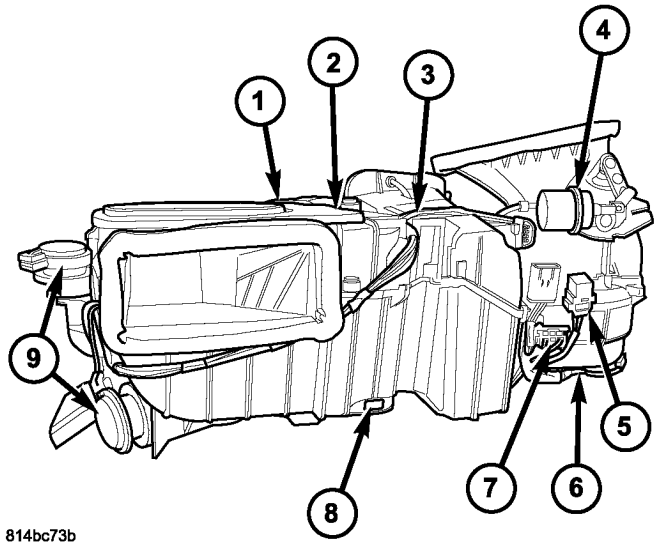

Fig. 7 HVAC Housing - LHD shown, RHD similar

1 - HVAC HOUSING
2 - HEATER CORE
3 - A/C EVAPORATOR
4 - RECIRCULATION DOOR ACTUATOR
5 - BLOWER MOTOR RELAY
6 - BLOWER MOTOR
7 - BLOWER MOTOR RESISTOR
8 - BLEND DOOR ACTUATOR
9 - MODE DOOR ACTUATORS

The heating-A/C system is a blend-air type system. The blend-air door controls the amount of conditioned air that is allowed to flow through, or around, the heater core.

The A/C system is designed for the use of a non-CFC, R-134a refrigerant and uses an A/C evaporator to cool and dehumidify the incoming air prior to blending it with the heated air. A temperature control determines the discharge air temperature by operating the blend door actuator, which moves the blend-air door. This allows an almost immediate control of the output air temperature of the system. The two mode door actuators operate the mode-air doors which direct the flow of the conditioned air out the various air outlets, depending on the mode selected. The recirculation door actuator operates the recirculation-air door which closes off the fresh air intake and recirculates the air already inside the vehicle. The electric recirculation door actuator is connected to the vehicle electrical system by the HVAC wire harness and, the recirculation door actuator and mode door actuators are connected to the A/C-heater control by the HVAC vacuum harness. The blower motor controls the velocity of air flowing through the HVAC housing assembly by spinning the blower wheel within the HVAC housing at the selected speed by use of the blower motor resistor.

The HVAC housing must be removed from the vehicle and disassembled for service of the heater core, A/C evaporator, blend-air and mode-air doors. The air inlet housing must be removed from HVAC housing and disassembled for service of the recirculation-air door.

REMOVAL

HVAC HOUSING

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. This is the only sure way to disable the airbag system. Failure to take the proper precautions could result in an accidental airbag deployment and possible personal injury or death.

NOTE: The HVAC housing must be removed from the vehicle and the two halves of the housing separated for service of the heater core, A/C evaporator, blend door and each of the various mode doors.

(1) Disconnect and isolate the negative battery cable.